

Novel graph modeling framework for feature importance determination in unsupervised learning

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Abstract. Not all features are created equal. Hence, feature importance determination is one of the most fundamental problems in Machine Learning. Most feature importance methods rely on the existence of a target feature (response, output, y) to understand the importance of each feature. But if the target feature is not present and we only have the independent features, then only unsupervised methodologies can be applied; feature importance is not easy to calculate [6]. In this article, a novel graph model methodology is proposed to identify the feature importance of datasets without a target feature. In the Proposed Ranking Approach, the target feature columns of 7 datasets were hidden, and their independent feature importance was calculated utilizing the Proposed Ranking Approach and another unsupervised method (Gower Distance) [4]. After that, the feature importance was calculated using the full data set (with the target feature) using commonly used supervised methods (Random Forest and Decision Tree algorithms). Finally, the rank of the most impactful features was compared for all methodologies [5]. It was concluded that each algorithm delivered a different feature importance rank that may or may not match with the other's output, but all the approaches have some level of overlap feature importance order. For this approach, the Python graph module (NetworkX and PageRank) and UCI machine learning datasets were used (Machine Learning Repository 2018).

1 Introduction

The feature importance problem has been extensively studied in the context of supervised learning [2, 7, 12, 8, 15, 19, 17, 18] but relatively few works have been published in the unsupervised approach [6, 3]. One of the main applications of determining feature importance is to perform feature selection, mainly in a supervised context. This is necessary when you have too many features for

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the number of samples [14], or when it is necessary to improve modeling performance by removing colinear features. Feature selection based on feature importance determination is a recurrent procedure in order to reduce the processing time of any modeling approach. In an unsupervised learning context, feature importance methodologies are applied to aid feature selection to find the smallest feature subset that best uncovers “interesting natural” groupings (or, clusters) from data according to the chosen criterion [6, 9]. There are no approaches that can identify the feature importance in an unsupervised learning problem. The clustering model takes the entire dataset and identifies the separability of subsets of the population who shared the same relations. However, clustering methods are unable to identify the order of importance which can help to reduce the number of features being analyzed. In cluster techniques, the records in a dataset are organized into different logical groupings. The data are grouped in such a way that records inside the same group are more similar than records outside the group, and clustering automatically split the dataset into groups based on their similarity matrixes (e.g., similarity, Manhattan distance) [9]. Due to the logical grouping used in clustering methodologies, information regarding feature importance cannot be extracted. Dealing with a large number of dimensions and data items is time-consuming. For future iterations all columns need to be analyzed, due to lack of ranking. Effectiveness of the clustering method depends on the definition of “distance”. For more on distance-based clustering techniques, please refer to (author?) [11, 16, 1]. If an obvious distance measure does not exist, one must “define” it, which is not always easy, especially in multidimensional spaces. The framework proposed in this article attempts to bridge the gap between supervised and unsupervised feature selection approaches in graph learning. In the Proposed Ranking Approach, the ranking of features is achieved by an iterative process of determining the rank of each feature by calculating the density of the node on a directed graph using Google PageRank (explained on section 3. Proposed Ranking Approach) [13]. [10] has proposed a methodology to calculate the importance of features in an undirected graph using PageRank but the order is already pre-established with the help of subspace preference clusters in order to define the edges’ linking features.

In this paper, a novel graph methodology is proposed for feature selection for unsupervised models. Briefly, the approach is as follows:

- 1) Convert all the columns and rows as nodes and edges into a directed graph.
- 2) Calculate the rank of each node by using the Google PageRank algorithm [13].
- 3) Set the highest ranked node as the starting node in the graph and build the graph again.
- 4) This process iterates n -times, n being the number of independent features and creates a matrix of size $m \times n$ (m = number of nodes, n = number of iterations) as an output of each iteration.
- 5) Calculate the minimum score of each node out of all iterations and sum the score of the all the nodes at the attribute level which are created in Step [1].
- 6) Sort by ascending score value order (Figure 1).

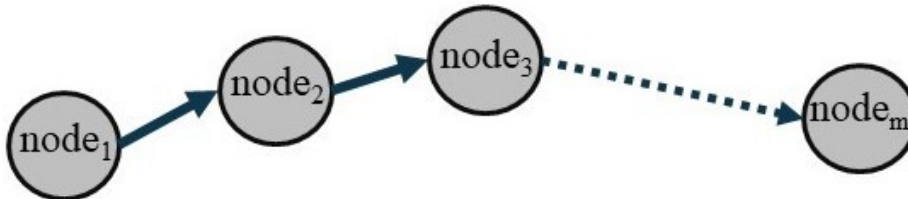


Figure 1. Basic elements of a directed graph: nodes and directed arrows.

2 Proposed Ranking Approach

2.1 Building the graph for given dataset

In this paper, 7 datasets from UCI machine learning datasets (Machine Learning Repository 2018) were utilized to run 4 approaches: 1) the Proposed Ranking Approach, 2) Random Forest, 3) Decision Tree and, 4) Clustering (using Gower Distance). The internal default parameters for Random Forest and Decision Tree were used. The columns of the seven datasets were converted into dummy columns (including base) and it was assumed that the attributes are categorical in nature. In the absence of categorical columns, continuous attribute/s is/are treated as categorical and dummy columns. By default, directed edges are in the sequence in which columns appear in the dataset. Using this approach, a directed graph can be created on any type of data set which is numeric and categorical in nature. Response features were not included in the graph as this is an application in unsupervised learning.

2.2 Calculate the rank of nodes

- A. Initially, a graph is built with the unsorted attribute order.
- B. Google PageRank algorithm was applied, and the highest scored node was computed (dummy column/feature from 3.1).
- C. The highest scored column/feature was saved.
- D. The highest scored column/feature was removed (C) and the rank of remaining node for next highest scored feature was re-computed and the process was iterated until the last node and the score was computed until the last feature and stored in matrix (F).
- E. The highest ranked feature (C) is set as starting node in graph with no change in the order of other features and the same steps are repeated from step (B) to step (D).
- F. Finally, the output was saved as matrix of $m \times n$ (m = number of nodes, n = number of iteration) from step (E).
- G. The minimum score of each dummy features out of all iterations is stored.
- H. The sum of each group of dummies at each feature level from step (G) is calculated (see Table 4).
- I. The output of the model (features) was sorted in ascending order and ranked.

3 Experimental results

3.1 Identification of feature importance (rank): Supervised learning vs Proposed Ranking approach

The feature ranking results of 3 of the 7 experiments using two supervised approaches and the Proposed Ranking approach are shown below. Refer to the Appendix section for the remaining feature ranking results. The ranking results are shown in a heatmap table where the most important feature is labeled “1” (dark green) and the least important feature is labeled in red. The heat map coloration gradient is shown in the figure below.



109 Titanic dataset [\[link\]](#):

110 Table 1. Ranking results from Random Forest, Decision Trees, and Proposed Ranking methodology
 111 experiments using the Titanic dataset. In the table below, the top 6 features identified by each of
 112 the three algorithms are shown.

feature of dataset	RF Rank	DT Rank	PR Rank
sex	1	1	1
Pclass	2	2	2
Age_prect_level	4	4	3
Fare_prect_level	3	5	4
Parent Child	6	6	5
Sib Spou	5	3	6

113
 114 RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach
 115

116 Iris dataset [\[link\]](#):

117 Table 2. Ranking results from Random Forest, Decision Trees, and Proposed Ranking methodology
 118 experiments using the Iris dataset. In the table below, the top 4 features identified by each of the
 119 three algorithms are shown.

feature of dataset	RF Rank	DT rank	PR rank
petal_width	1	1	1
sepal_length	3	3	2
sepal_width	4	4	3
petal_length	2	2	4

120
 121 RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach
 122

123 Breast Cancer dataset [\[link\]](#):

124 Table 3. Ranking results from Random Forest, Decision Trees, and Proposed Ranking Approach
 125 experiments using the Breast Cancer dataset. In the table below, the top 9 features identified by
 126 each of the three algorithms are shown.

feature of dataset	RF Rank	DT rank	PR rank
HOMA	5	2	1
Glucose	1	1	2
Age	2	3	2
BMI	4	2	3
Adiponectin	8	6	4
Resistin	3	5	5
Insulin	7	4	6
MCP.1	9	6	7
Leptin	6	4	8

RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach

Each of the three algorithms returns a different feature rank order that might not match in its entirety. But all the algorithms have some level of rank overlap, and the Proposed Ranking methodology was able to identify the top ranked features in all the experiments.

3.2 Clustering of the resulting top ranked features sets – comparing separability power of approaches

In this section a clustering algorithm was used to compare the Proposed Ranking methodology and Random Forest results from section 3.1 regarding their fitness of separability using a clustering algorithm (Gower 1971) where the Gower Distance is used to calculate distance (in this paper the Daisy R-Package was used) between two entities whose attributes have a mix of categorical and numerical values.

5 sets of clustering experiments were performed in which the datasets used were 3 sets of the top ranked features from section 3.1 (Tables 1-3).

The silhouette width displays a measure of how close each point in one cluster is to points in the neighboring clusters and thus provides a way to assess parameters like number of clusters visually. This measure has a range of $[-1, 1]$. Silhouette coefficients (as these values are referred to as) near +1 indicate that the sample is far away from the neighboring clusters. A value of 0 indicates that the sample is on or very close to the decision boundary between two neighboring clusters – indicating poor separability. Negative values indicate that those samples might have been assigned to the wrong cluster (scikit-learn developers 2021). From visually inspecting the Silhouette plots (fig 1-3), a value of 0.2 was selected as the threshold of good separability; 0.20 is the minimum threshold for homogeneous clusters.

Titanic dataset [\[link\]](#):

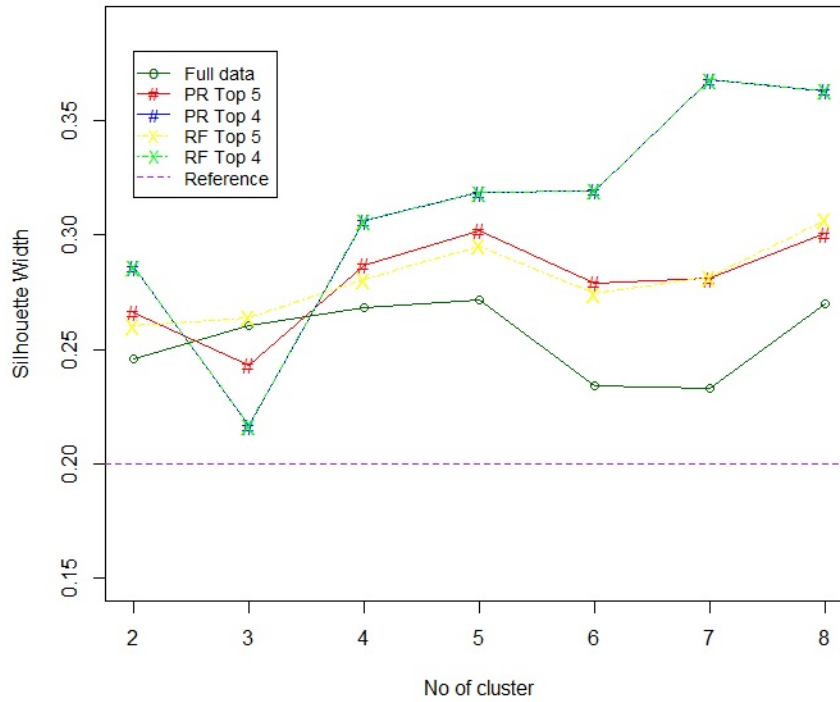


Figure 1: Silhouette width plot versus the number of clusters in each clustering experiment run with different features on the “Titanic” dataset. Using all features plotted in green (o), using the top 5 features of Proposed Ranking approach in red (#), top 4 features of Proposed Ranking approach in blue (#), top 5 features of Random Forest in yellow (x), top 4 features of Random Forest in green (x). The horizontal dashed line represents the benchmark threshold of a Silhouette width of 0.2 which is the threshold for “good” separability, used to select the optimal number of clusters: above 0.2 the model has a good separability, below a bad separability. The further away from this threshold, the better the model separability is.

Breast Cancer dataset [\[link\]](#):

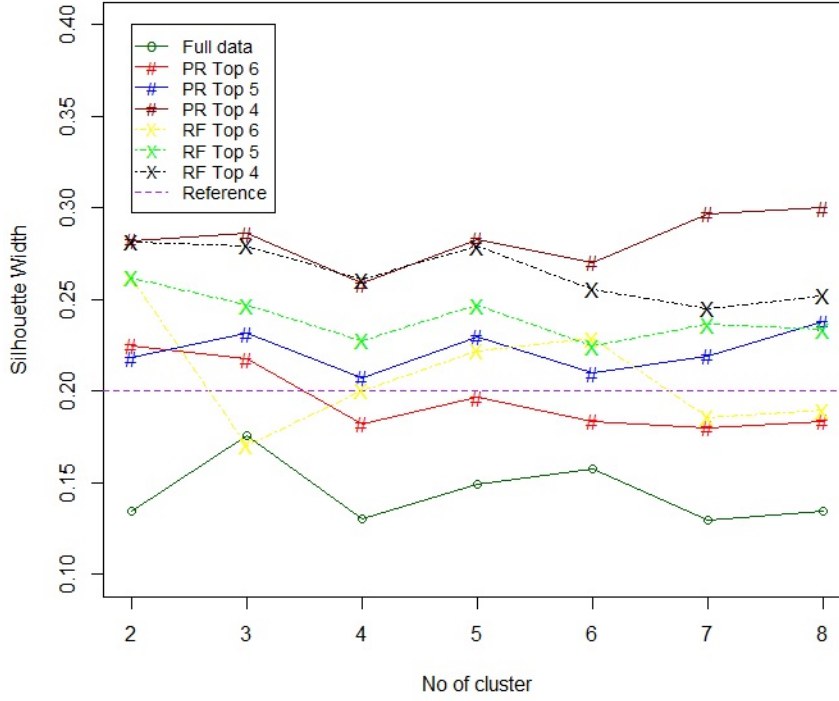


Figure 2: Silhouette width plot versus the number of clusters in each experiment run with different features on the "Breast Cancer" dataset. Clustering approach results using all features plotted in green (o), using the top 6 features of Proposed Ranking approach in red (#), top 5 features of Proposed Ranking approach in blue (#), top 4 features of Proposed Ranking approach in dark red (#), top 6 features of Random Forest in yellow (x), top 5 features of Random Forest in green (x), top 4 features of Random Forest in black (x). The horizontal dashed line represents the benchmark threshold of a Silhouette width of 0.2 which is the threshold for "good" separability, used to select the optimal number of clusters: above 0.2 the model has a good separability, below a bad separability. The further away from this threshold, the better the model separability is.

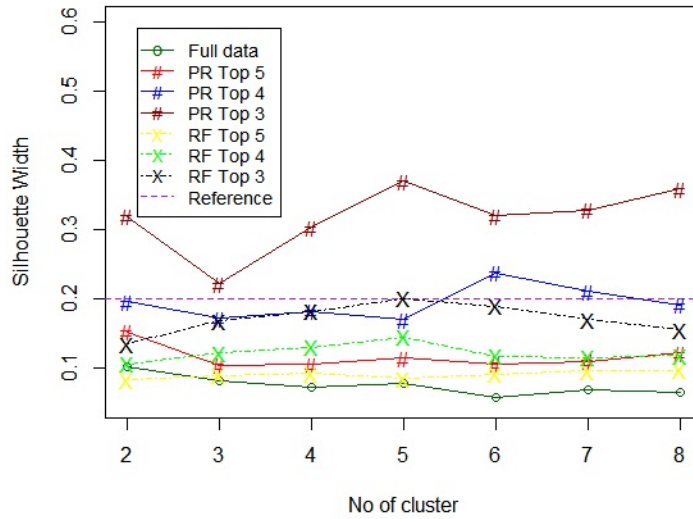


Figure 3: Silhouette width plot versus the number of clusters in each experiment run with different features on the “Immunotherapy” dataset. Clustering approach results using all features plotted in green (o), using the top 5 features of Proposed Ranking approach in red (#), top 4 features of Proposed Ranking approach in blue (#), top 3 features of Proposed Ranking approach in dark red (#), top 5 features of Random Forest in yellow (x), top 4 features of Random Forest in green (x), top 3 features of Random Forest in black (x). The horizontal dashed line represents the benchmark threshold of a Silhouette width of 0.2 which is the threshold for “good” separability, used to select the optimal number of clusters: above 0.2 the model has a good separability, below a bad separability. The further away from this threshold, the better the model separability is.

Chess dataset [\[link\]](#) :

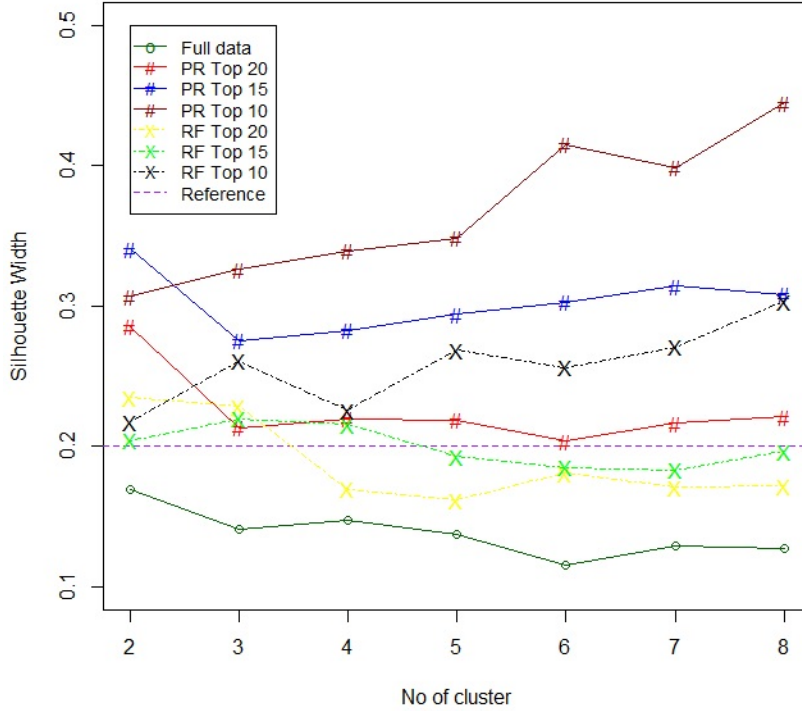


Figure 4: Silhouette width plot versus the number of clusters in each experiment run with different features on the “Chess” dataset. Clustering approach results using all features plotted in green (o), using the top 20 features of Proposed Ranking approach in red (#), top 15 features of Proposed Ranking approach in blue (#), top 10 features of Proposed Ranking approach in dark red (#), top 20 features of Random Forest in yellow (x), top 15 features of Random Forest in green (x), top 10 features of Random Forest in black (x). The horizontal dashed line represents the benchmark threshold of a Silhouette width of 0.2 which is the threshold for “good” separability, used to select the optimal number of clusters: above 0.2 the model has a good separability, below a bad separability. The further away from this threshold, the better the model separability is.

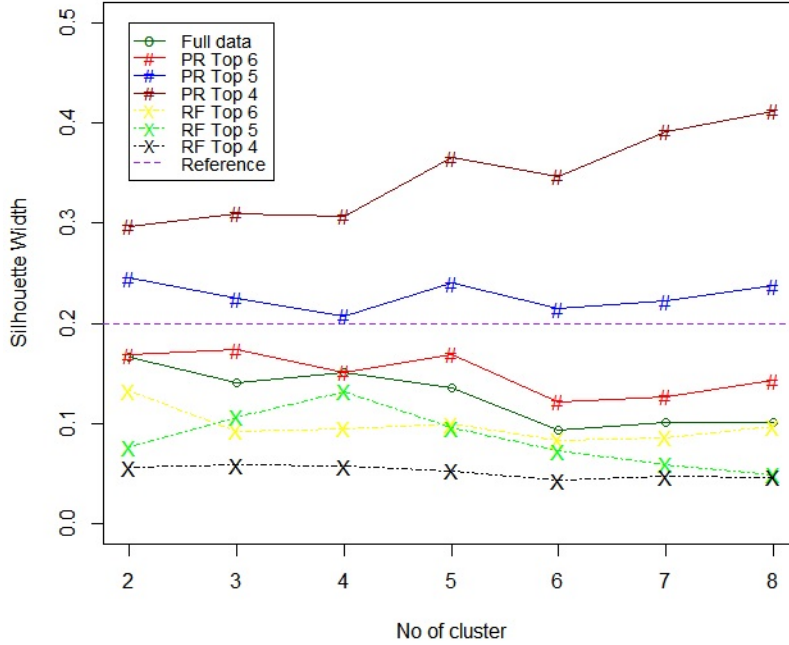


Figure 5: Silhouette width plot versus the number of clusters in each experiment run with different features on the Metro” Interstate Traffic Volume” dataset. Clustering approach results using all features plotted in green (o), using the top 6 features of Proposed Ranking approach in red (#), top 5 features of Proposed Ranking approach in blue (#), top 4 features of Proposed Ranking approach in dark red (#), top 6 features of Random Forest in yellow (x), top 5 features of Random Forest in green (x), top 4 features of Random Forest in black (x). The horizontal dashed line represents the benchmark threshold of a Silhouette width of 0.2 which is the threshold for “good” separability, used to select the optimal number of clusters: above 0.2 the model has a good separability, below a bad separability. The further away from this threshold, the better the model separability is.

From visually inspecting Figures 2-5, the Proposed Ranking Approach systematically produced top-ranked feature segments (from section 3.1) that when used in a clustering approach allowed for a better separability than when using the top-ranked feature segments from the Random Forest approach. Silhouette width Coefficients of the cluster experiments using the Proposed Ranking Approach top features were always higher than when using the top features from the Random Forest approach.

4 Conclusion

In this paper, a novel methodology was proposed to assess the feature importance in unsupervised learning, using a novel graph application of Google PageRank. The fitness of the feature importance determination on 7 different experiments was assessed. Its effectiveness was compared

221 against the benchmark unsupervised and supervised feature selection methods. With respect to
222 known methods like supervised (Random Forest and DT) and unsupervised (clustering), the model
223 performance confirms the ability of the Proposed Ranking approach for feature selection. The
224 Proposed Ranking Approach proves that without target features, the Proposed Ranking Approach
225 can identify the feature important [section 3.1] as well identified feature also provided the better
226 clustering using same feature importance [section 3.2] While the emphasis of this paper was on
227 estimating feature importance in unsupervised learning, it is worth mentioning that the idea of
228 graph theory-based feature importance measure may also be extended to semi-supervised learning
229 feature.

230 5 GitHub link

231 [Source code](#)

232 6 Conflict of interests

233 This work is based on personal experiments, research, and publicly available datasets and is neither
234 linked to Microsoft or Bayer's data in any way.

235 7 Appendix

236 Comparing the feature importance of supervised algorithms versus the Proposed Ranking Ap-
237 proach.

238 Chess dataset [\[link\]](#) :

239 Table 4: Ranking results from Random Forest, Decision Trees, and Proposed Ranking Approach
240 experiments using the Breast Cancer dataset. In the table below, the top 36 features identified by
241 each of the three algorithms are shown.

feature of dataset	RF Rank	DT rank	PR rank
hdchik	33	12	1
wkpos	5	11	2
wkovl	12	12	3
wknck	2	2	4
wkna8	6	4	5
wkcti	20	12	6
thrsk	30	12	7
skrxp	14	9	8
rxmsq	18	12	9
simpl	22	12	10
skach	34	12	11
mulch	13	8	12
r2ar8	11	12	13
reskr	26	12	14
rkxwp	21	12	15
wtoeg	25	12	16
bklbk	10	6	17
dwipd	23	12	18
stlmt	27	10	19
blnwy	19	12	20
dsopp	32	12	21
nimmx	1	3	22
cntxt	15	12	23
bxqsq	3	1	24
blxwp	17	12	25
spcop	36	12	26
bkxwp	8	12	27
qxmsq	31	12	28
bkxcr	9	12	29
bkxbq	7	5	30
bkspr	16	12	31
reskd	35	12	32
skewr	28	12	33
bkona	24	12	34
bkon8	29	12	35
katri	4	7	36

243 RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach

244 Immunotherapy dataset [\[link\]](#) :

245 Table 5: Ranking results from Random Forest, Decision Trees, and Proposed Ranking Approach
 246 experiments using the Breast Cancer dataset. In the table below, the top 7 features identified by
 247 each of the three algorithms are shown.

feature of dataset	RF Rank	DT rank	PR rank
Sex	7	2	1
Type	6	6	2
induration diameter	3	3	3
Number_of_Warts	2	5	4
Age	5	1	5
Time	1	3	6
Area	4	4	7

248
 249 RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach

250 Traffic dataset [\[link\]](#) :

251 Table 6: Ranking results from Random Forest, Decision Trees, and Proposed Ranking Approach
 252 experiments using the Breast Cancer dataset. In the table below, the top 8 features identified by
 253 each of the three algorithms are shown.

feature of dataset	RF Rank	DT rank	PR rank
clouds_all_level	6	3	1
temp_level	5	7	2
holiday	8	8	3
weather_main	7	6	4
Month	3	5	5
Hour	1	1	6
day	2	2	7
weather_description	4	4	8

254
 255 RF : Random Forest, DT : Decision Tree, PR : Proposed Ranking Approach

256 Table 7: Calculation of the rank (step “H” of section 2.2)

		iteration weight (total run = 5)						Min all Iteration	Sum of individual variable	Order of Impotance
Column1		0	1	2	3	4	5			
attribute level	Age_prect_level_1	0.111	0.104	0.143	0.078	0.071	0.077	0.071	0.286	Age-3
	Age_prect_level_2	0.111	0.104	0.143	0.078	0.071	0.077	0.071		
	Age_prect_level_3	0.104	0.097	0.143	0.078	0.071	0.077	0.071		
	Age_prect_level_4	0.092	0.086	0.143	0.078	0.071	0.077	0.071		
	Fare_prect_level_1	0.143	0.111	0.104	0.143	0.072	0.085	0.072	0.346	Fare-4
	Fare_prect_level_2	0.143	0.111	0.104	0.143	0.072	0.085	0.072		
	Fare_prect_level_3	0.143	0.111	0.140	0.143	0.116	0.138	0.111		
	Fare_prect_level_4	0.143	0.111	0.104	0.143	0.091	0.104	0.091		
	Parent_Child_0	0.138	0.129	0.119	0.100	0.071	0.059	0.059	0.423	Parent_child-5
	Parent_Child_1	0.127	0.115	0.103	0.085	0.071	0.059	0.059		
	Parent_Child_2	0.148	0.134	0.118	0.097	0.071	0.059	0.059		
	Parent_Child_3	0.130	0.120	0.108	0.090	0.071	0.059	0.059		
	Parent_Child_4	0.097	0.086	0.074	0.072	0.071	0.059	0.059		
	Parent_Child_5	0.097	0.086	0.074	0.072	0.071	0.059	0.059		
	Parent_Child_6	0.127	0.110	0.094	0.091	0.071	0.128	0.071		
	Pclass_1	0.143	0.105	0.143	0.143	0.071	0.072	0.071	0.214	Pclass-2
	Pclass_2	0.143	0.105	0.143	0.143	0.071	0.072	0.071		
	Pclass_3	0.143	0.105	0.143	0.143	0.071	0.072	0.071		
	sex_0	0.116	0.111	0.088	0.078	0.086	0.077	0.077	0.154	Sex-1
	sex_1	0.116	0.111	0.088	0.078	0.086	0.077	0.077		
	Sib_Spou_0	0.095	0.087	0.095	0.095	0.067	0.077	0.067	0.484	Sib_spou-6
	Sib_Spou_1	0.095	0.087	0.095	0.095	0.067	0.077	0.067		
	Sib_Spou_2	0.095	0.087	0.095	0.095	0.067	0.077	0.067		
	Sib_Spou_3	0.095	0.087	0.095	0.095	0.067	0.077	0.067		
	Sib_Spou_4	0.119	0.111	0.119	0.119	0.073	0.077	0.073		
	Sib_Spou_5	0.119	0.111	0.119	0.119	0.073	0.077	0.073		
	Sib_Spou_8	0.119	0.111	0.119	0.119	0.073	0.077	0.073		

8

Acknowledgements

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